

Tilt2PC — USB Connection Guide

Low-latency Android tilt controller for Asphalt 9

Why USB Mode?

Using USB tethering gives:

-  Lower latency
 -  More stable steering
 -  Faster nitro response
 -  Less packet loss
 -  No WiFi/router delay
 -  Better overall gameplay feel
-

Requirements

Before starting, make sure you have:

- Android phone
 - USB data cable
 - Python installed on PC
 - Tilt2PC APK installed on phone
 - `server.py` file on PC
-

Recommended Folder Structure

Create this folder on your PC:

```
Desktop/  
└─ tilt-server/  
    └─ server.py
```



STEP 1 — Connect Phone to PC

Connect your Android phone using a USB cable.

On the phone:

```
USB Mode → File Transfer
```

⚠ Use a proper data cable. Charging-only cables may not work.



STEP 2 — Enable USB Tethering

On Android:

```
Settings  
→ Hotspot & Tethering  
→ USB Tethering  
→ ON
```

Keep USB tethering enabled while playing.



STEP 3 — Find PC USB IP Address

Open Windows CMD or PowerShell.

Run:

```
ipconfig
```

Look for:

```
Ethernet adapter Ethernet 2
```

OR:

Remote NDIS

Find:

IPv4 Address

Example:

10.87.111.183

⚠ IMPORTANT: Do NOT use:

- localhost
- 127.0.0.1
- old WiFi IP

Use ONLY the USB tethering IP.



STEP 4 — Start Python Server

Open terminal inside:

Desktop/tilt-server

Run:

```
python server.py
```

Expected output:

```
=====
Tilt2PC Server
=====
IP: 10.x.x.x
PORT: 3333

Waiting for phone...
```



STEP 5 — Connect the Android App

Open the Tilt2PC APK.

Enter:

Field	Value
IP	Your USB tethering IP
PORT	3333

Example:

```
IP: 10.87.111.183
PORT: 3333
```

Then press:

```
CONNECT
```



Successful Connection

If connection succeeds:

Phone app shows:

```
Connected
```

PC terminal shows:

```
[CONNECTED]
```

Controls

Action	Function
Tilt Left	Steer Left
Tilt Right	Steer Right
Nitro Button	Nitro
Drift Button	Drift
Brake Button	Brake

Common Problems

App says "Disconnected"

Fixes:

- Check USB tethering is ON
- Recheck IP using `ipconfig`
- Ensure server.py is running
- Ensure port is `3333`
- Uninstall old APK and install latest APK

Wrong IP

USB tethering IP changes sometimes.

Always verify using:

```
ipconfig
```

before connecting.

Server not starting

Install dependencies:

```
pip install websockets pynput
```

Current Architecture

```
Android APK
  ↓
USB Tethering
  ↓
Python WebSocket Server
  ↓
Keyboard Input
  ↓
Asphalt 9
```

Future Improvements

Planned upgrades:

-  Virtual Xbox controller support
-  UDP networking
-  Analog steering
-  Even lower latency
-  Better drift handling

Notes

- Keep phone connected while playing
- Keep server terminal open
- USB mode gives much smoother gameplay than WiFi
- APK mode is faster than Expo Go

Enjoy Driving

Tilt2PC is now running locally with low-latency USB communication for Asphalt 9.